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Kelas : A

1. Codengan perkalian array 3D dengan deklarasi variabel array

```
1 def array_multiplication_3d(A, B):
2     if len(A) != len(B):
3         raise ValueError("Array sizes do not match!")
4     if len(A[0]) != len(B[0]):
5         raise ValueError("Array sizes do not match!")
6     if len(A[0][0]) != len(B[0][0]):
7         raise ValueError("Array sizes do not match!")
8     result = []
9     for i in range(len(A)):
10        layer = []
11        for j in range(len(A[i])):
12            row = []
13            for k in range(len(A[i][j])):
14                row.append(A[i][j][k] * B[i][j][k])
15            layer.append(row)
16        result.append(layer)
17    return result
18
19 def display(array):
20     for i, layer in enumerate(array):
21         print(f"Layer {i + 1}:")
22         for row in layer:
23             print(row)
24         print()
25
26 def main():
27     print("=== 3D ARRAY MULTIPLICATION ===")
28
29     A = [
30         [
31             [1, 2],
32             [3, 4]
33         ],
34         [
35             [5, 6],
36             [7, 8]
37         ]
38     ]
39
40     B = [
41         [
42             [2, 3],
43             [4, 5]
44         ],
45         [
46             [6, 7],
47             [8, 9]
48         ]
49     ]
50
51     print("\nArray A:")
52     display(A)
53     print("Array B:")
54     display(B)
55     result = array_multiplication_3d(A, B)
56     print("The product of A x B:")
57     display(result)
58
59 if __name__ == "__main__":
60     main()
```

2. Codengan perkalian array 3D dengan input user

```
1 def input_array_3d(name, layers, rows, cols):
2     array = []
3
4     print(f"\nEnter elements for Array {name}:")
5
6     for i in range(layers):
7         layer = []
8
9         print(f"Layer {i + 1}:")
10
11        for j in range(rows):
12            row = []
13
14            for k in range(cols):
15                value = int(input(f"Element [{i}][{j}][{k}]: "))
16                row.append(value)
17
18            layer.append(row)
19
20        array.append(layer)
21
22    return array
23
24 def array_multiplication_3d(A, B):
25     if len(A) != len(B):
26         raise ValueError("Array sizes do not match!")
27
28     if len(A[0]) != len(B[0]):
29         raise ValueError("Array sizes do not match!")
30
31     if len(A[0][0]) != len(B[0][0]):
32         raise ValueError("Array sizes do not match!")
33
34     result = []
35
36     for i in range(len(A)):
37         layer = []
38
39         for j in range(len(A[i])):
40             row = []
41
42             for k in range(len(A[i][j])):
43                 row.append(A[i][j][k] * B[i][j][k])
44
45             layer.append(row)
46
47         result.append(layer)
48
49     return result
50
51 def display(array):
52     for i, layer in enumerate(array):
53         print(f"Layer {i + 1}:")
54         for row in layer:
55             print(row)
56         print()
57
58 def main():
59     print("=== 3D ARRAY MULTIPLICATION ===")
60
61     layers = int(input("Enter number of layers: "))
62     rows = int(input("Enter number of rows: "))
63     cols = int(input("Enter number of columns: "))
64
65     A = input_array_3d("A", layers, rows, cols)
66     B = input_array_3d("B", layers, rows, cols)
67
68     print("\nArray A:")
69     display(A)
70
71     print("Array B:")
72     display(B)
73
74     result = array_multiplication_3d(A, B)
75
76     print("The product of A x B:")
77     display(result)
78
79
80 if __name__ == "__main__":
81     main()
```

3. Output untuk program 1 (main.py)

```
=== 3D ARRAY MULTIPLICATION ===

Array A:
Layer 1:
[1, 2]
[3, 4]

Layer 2:
[5, 6]
[7, 8]

Array B:
Layer 1:
[2, 3]
[4, 5]

Layer 2:
[6, 7]
[8, 9]

The product of A x B:
Layer 1:
[2, 6]
[12, 20]

Layer 2:
[30, 42]
[56, 72]
```

4. Output untuk program 2 (main2.py)

```
=== 3D ARRAY MULTIPLICATION ===
Enter number of layers: 2
Enter number of rows: 2
Enter number of columns: 2

Enter elements for Array A:
Layer 1:
Element [0][0][0]: 1
Element [0][0][1]: 3
Element [0][1][0]: 7
Element [0][1][1]: 2
Layer 2:
Element [1][0][0]: 1
Element [1][0][1]: 4
Element [1][1][0]: 1
Element [1][1][1]: 3

Enter elements for Array B:
Layer 1:
Element [0][0][0]: 3
Element [0][0][1]: 1
Element [0][1][0]: 1
Element [0][1][1]: 2
Layer 2:
Element [1][0][0]: 1
Element [1][0][1]: 2
Element [1][1][0]: 3
Element [1][1][1]: 4

Array A:
Layer 1:
[1, 3]
[7, 2]
Layer 2:
[1, 4]
[1, 3]

Array B:
Layer 1:
[3, 1]
[1, 2]
Layer 2:
[1, 2]
[3, 4]

The product of A x B:
Layer 1:
[3, 3]
[7, 4]
Layer 2:
[5, 8]
[3, 12]
```

5. Programnya

https://github.com/aldyansyahombi/matrix3d_pcd